Learning Domain-Invariant Model for WiFi-Based Indoor Localization

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***Abstract*—WiFi-based indoor localization has gained widespread attention due to the pervasive availability of WiFi Access Points (APs). While signal processing-based methods can achieve decimeter-level localization, their performance is constrained by the limited spatial resolution of WiFi systems, especially in complex environments with strong interference. By contrast, deep learning-based methods have achieved impressive performance even in complex environments, which however often fail to generalize to new environments. In this paper, we propose a novel framework to learn domain-invariant model for WiFi-based indoor localization, which maintains impressive performance across different environments. The key insight is to design a deep learning-based WiFi localization system through the perspective of signal processing. Specifically, we let the neural network estimate APs-centered polar coordinates to avoid fitting the coordinates of APs strongly correlated with the environment, enabling us to obtain the domain-invariant model. To unleash the potential of neural networks in regressing high-precision parameters, we design a beamforming layer to integrate the knowledge of signal processing. Furthermore, we propose a multi-task learning scheme to further improve localization accuracy. Extensive experiments on diverse datasets have demonstrated that the localization performance of our method outperforms state-of-the-art methods and demonstrates superiority under cross-domain conditions.**

***Index Terms*—Channel state information, deep learning, indoor localization, WiFi.**

I. Introduction

INDOOR localization has been an exciting area over the past two decades [1], with various practical applications including security surveillance [2], indoor navigation [3], and retail business [4].

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Due to the ubiquitous presence of WiFi Access Points (APs) [5], [6], WiFi-based systems [7], [8] have attracted greater attention than radar-based systems [9], [10]. Recently, the essence of WiFi-based localization systems is to use Channel State Information (CSI) between the clients and APs for localization [11].

One straightforward solution is first to estimate the Angle of Arrival (AoA) and Time of Flight (ToF) information from CSI through various signal processing algorithms, and then perform triangulation based on the coordinates of APs [12], [13], [14], [15]. The signal processing-based methods rely on the domain-invariant mathematical models and thus can be deployed in any environment. Nonetheless, the limited spatial resolution of commercial WiFi devices constrains the performance of these algorithms, subsequently restricting the localization performance [16]. What is worse, the performance of these methods deteriorates significantly in complex environments with strong interference [17].

Another approach is to estimate the coordinates of clients with the deep neural network directly. Deep learning-based methods extract the characteristics of CSI at different locations from a vast amount of data, which is utilized to tune the neural network parameters. Models established through this approach exhibit superior performance compared to signal processing-based methods, particularly in complex environments with strong interference [17]. Unfortunately, existing deep learning-based methods tend to overfit the features within the training domain, which suffers from severe performance degradation when deployed in new domains.

In this paper, we integrate the cross-domain capability of signal processing and the fitting ability of deep learning techniques for high-precision parameter estimation and reliable localization. The key insight of the proposed method lies in two aspects:

First, existing deep learning-based methods fit the coordinates of APs that are strongly correlated with the environment into network parameters. Due to the complexity of indoor environments, the coordinates of APs in different scenarios are usually different, as Fig. 1 shows. This makes the trained models highly dependent on the environment. Distinguished from previous works, we propose using APs-centered polar coordinates as the outputs of the network. Our observation is that the relative spatial relationship between APs and clients remains consistent despite the change of environments, and APs-centered polar coordinates encapsulate this characteristic.

Second, the regression precision of polar coordinates can be enhanced by incorporating the knowledge of signal processing into deep learning. Specifically, we observe that the fundamental

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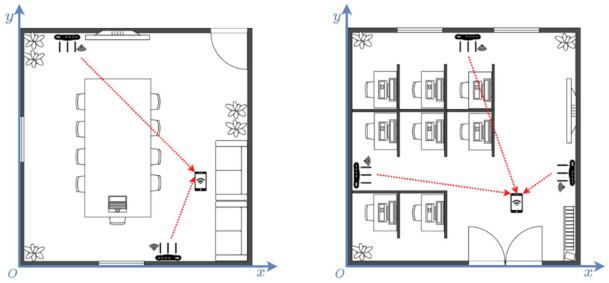

(a) Meeting Room (b) Office

Fig. 1. Cross-domain localization, where the coordinates of APs in different scenarios are usually different due to the complexity of indoor environments.

principle of beamforming and the forward propagation of fully connected layers are essentially the same. Based on this observation, we can implement beamforming with a fully connected layer to further unleash the potential of neural networks for high-precision parameter estimation.

To further improve the localization performance, we propose a multi-task learning scheme for localization. This scheme enables the model to learn features of Cartesian coordinates and polar coordinates simultaneously, leading to high-precision localization accuracy and good generalization ability.

Our main contributions can be summarized as follows.

- We propose to unleash the potential of neural networks for high-precision polar coordinates estimation with a beamforming layer. In this manner, we can integrate the advantages of signal processing and deep neural networks to learn a domain-invariant model.
- We propose a multi-task learning scheme for localization, which allows the neural network to simultaneously learn different representations for the signal from the same location, further improving localization accuracy.
- We conduct extensive experiments on multiple datasets across various indoor environments. The experimental results demonstrate that our proposed method significantly outperforms state-of-the-art signal processing-based approaches and maintains impressive cross-domain robustness compared with state-of-the-art deep learning-based methods. These datasets will be released to the research community and we believe these datasets will further advance the research of WiFi-based indoor localization.

The rest of this paper is as follows. Section II discusses related work and Section III presents the primer knowledge. In Section IV, we present the design of our method. The experimental settings and results are shown in Sections V and VI, respectively. Section VII demonstrates the performance of our approach in a large-scale environment. Section VIII draws the conclusions.

II. Related Work

A. Signal Processing-Based Methods

The commercial off-the-shelf WiFi devices are usually equipped with multiple antennas and can obtain physical channel measurements in subcarrier-level granularity [18], and thus various signal processing algorithms can be used to extract AoA and ToF information.

ToF-based methods: ToF-based methods require a large bandwidth to achieve a high ToF resolution [16]. With the limited bandwidth of WiFi devices, ToneTrack [14] gets a large virtual bandwidth by channel switching, and Chronos [19] gets accurate ToF estimations by frequency hopping. However, these methods can interrupt regular data communication which is not practical for WiFi-based localization. Different from the aforementioned works, SiFi [20] makes a particular observation: due to the frequency-locked-antenna design of Multiple-Input Multiple-Output (MIMO), only one value for the delay distortion that should cause all the direct-path ToFs of antennas to intersect at a single point. With this observation, SiFi achieves a median accuracy of 0.93 m while keeping data communication unaffected. Nonetheless, it needs to pull out antennas from a WiFi device to meet its theoretical model, which is also impractical. P2PLocate [21] separates robust CSI from backscatter, estimates fine-grained Doppler shift and coarse-grained ToF from it, and combines both to achieve single-antenna transmitter-receiver localization. [22] conducts extensive testing of the single-sided ranging protocol supported by Android 12 and confirms that the ranging results obtained from this method are not as accurate as those from Fine Time Measurement (FTM) ranging protocol. Furthermore, [22] proposes a series of related methods to enhance its performance. WiPeep [23] introduces the concept of non-cooperative WiFi localization and investigates its privacy implications. Specifically, WiPeep continuously moves one AP and injects fake data packets to elicit responses from the client. WiPeep formulates optimization equations by combining the ToF information obtained from response packets with the ground truth of AP movement trajectories, ultimately solving for the client's location.

AoA-based methods: ArrayTrack [12] achieves a high-precision localization performance by using the conventional MUSIC algorithm with 8 antennas, while it depends on the specific hardware that cannot be directly deployed on the commodity WiFi devices. Later, Ubicarse [24] and LTEye [25] use the idea of synthetic aperture to obtain a large array aperture, thereby improving the AoA estimation accuracy on commodity WiFi devices. However, this requires the person holding the device to perform a specific circular motion to achieve localization, which is not practical in many scenarios. SpotFi [13] successfully applies spatial smoothing MUSIC algorithm on commodity APs to jointly estimate AoA and ToF. While the spatial smoothing operation can avoid performance degradation caused by coherent signals, it will also sacrifice the array aperture and limit the spatial resolution [26]. RoArray [27] converts AoA estimation into a sparse recovery problem and can obtain a sharp AoA spectrum even in low signal-to-noise ratio (SNR) scenarios. It is worth noting that improper hyperparameter selection of RoArray will result in the miss detection of weak targets, and the computational complexity of RoArray is too high to achieve real-time localization. UbiLocate [28] utilizes Nelder-Mead search to achieve more accurate AoA estimation and combines it with ToF measurements with nanosecond-level precision to mitigate the performance degradation in Non-Line-of-Sight (NLoS) scenarios. Co-Loc [16] introduces a non-parametric metric to quantify the AoA estimation accuracy, and further improves the AoA estimation by leveraging the fact that the AoAs of multiple APs will intersect at one point. NLoc [29] is the first system that utilizes widely available multipath reflections to

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enable target localization even in the absence of the direct path without pre-training or fingerprinting. Based on the insight that the uncertain initial phase is constant once the RF oscillators are frequency locked, AutoLoc [30] develops a localization method that eliminates the need for calibrating uncertain initial phases. MultiLoc [31] proposes a localization system that integrates millimeter-wave and WiFi to achieve high accuracy and reliability. The localization system achieved a median error of only 0.18 m with off-the-shelf devices for the first time. [32] proposes a cost-effective and accurate method for localization with a high-speed RF mux and sub-packet switching to determine the AoA without the need for synchronization between the mux and the receiver, addressing the limitations in deploying phased antenna arrays for a large number of elements.

B. Deep Learning-Based Methods

Deep learning-based methods use the neural network to extract features of each location, and once trained, they can predict the location according to the input. The core of deep learning methods lies in the crafted input feature and well-designed network architecture.

Input Feature: To obtain robust features for complex environments and imperfect hardware systems, the input features should exhibit robustness to small-scale fading, system variations, and hardware impairments [33]. CiFi [34] utilizes CSI phase differences to estimate the AoA, which are then composed into images as inputs to deep convolutional neural networks [35]. [36] and [37] transform CSI into the AoA-ToF domain, aiming for the network to differentiate each location based on the rich multipath information encapsulated in these features. [38] extracts 8 different types of information from CSI as enhanced features for indoor localization and constructs the input feature through a data construction algorithm. To leverage prior works' knowledge of WiFi-based localization, DLoc [17] transforms the CSI of multiple APs through 2D Fast Fourier Transform (2D-FFT) into position heatmaps that reflect the client's location. Then, DLoc combines these heatmaps to create an image tensor as the input and converts the client's Cartesian coordinates label into an image with a Gaussian peak. LiPhi++ [39] proposes leveraging the sensing capability of transportable laser range scanners to automatically label WiFi scans, enabling data collection without the need for manual intervention.

Network Architecture: To distinguish different locations through input features, the network architecture needs to be carefully designed. DeepFi [40] adopts a greedy learning algorithm using a stack of Restricted Boltzmann Machines to train the deep neural network in a layer-by-layer manner. [41] explores the possibility of utilizing the temporal correlation of CSI features for localization based on a Long Short-Term Memory (LSTM) neural network. In [38], dual-attention mechanism deep neural network and bidirectional long short-term memory are introduced, and an attention mechanism is employed to extract features from CSI. [42] proposes an attention-augmented residual Convolutional Neural Network (CNN) to utilize the local information and global context in CSI. DLoc [17] leverages the network structures of image translation to process the input and output images. Additionally, DLoc incorporates a consistency decoder to eliminate ToF offsets. WePos [43] utilizes the pre-training techniques from the field of Natural Language Processing (NLP) to pre-train Received Signal Strength Indicator (RSSI) data from the APs of shopping malls. Then, through the integration of a pseudo-label acquisition algorithm to fine-tune the pre-trained model, Wepos achieves low-cost, high-precision, region-level indoor localization. By applying Model-Agnostic Meta-Learning (MAML) to fingerprint localization, MetaLoc [?] enables the model to quickly adapt to new environments based on the obtained meta-parameters, thus reducing human labor costs. Based on the domain adversarial neural network and a co-training-based semi-supervised learning framework, iToLoc [44] greatly mitigates the effect of signal variation, device heterogeneity, and database degradation for RSSI-based localization. MTLoc [45] utilizes the Multi-Target Domain Adaptive Network (MTDAN) to learn time-invariant, time-specific, and location-aware features from both the source and target domains, achieving automatic updates for the fingerprint database. Penetrative AI [46] even eliminates the need for designing and training their network, opting instead to utilize pre-trained large language models to process sensor information and obtain location information.

To sum up, the signal processing-based methods have strong cross-domain capabilities; nonetheless, their localization performance is limited by the spatial resolution of commercial WiFi devices, and degrades severely in strong interference environments. Although deep learning-based methods can achieve a high-precision localization performance in specific environments, they lack generalization capabilities. To overcome the limitations of these two types of methods, we use the neural network to regress APs-centered polar coordinates to integrate the cross-domain capabilities of signal processing and the fitting capabilities of deep learning.

III. Primer

A. Channel State Information

Assuming the transmitter has 1 antenna and the receiver is equipped with a uniform linear array (ULA) consisting of M antennas (the space interval between antennas is half-wavelength). For such a Single-Input Multiple-Output (SIMO) system, the transmitter transmits a set of OFDM signals over K subcarriers. The transmitted signals $\mathbf{s}$ can be written as:

$$\mathbf{s} = [s_1, s_2, \ldots, s_K]^T \tag{1}$$

where $(\cdot)^T$ is the transpose operator.

As the transmitted signals propagate through wireless channel $\mathbf{H}$, the received signals $\mathbf{r}$ can be written as:

$$\mathbf{r} = \mathbf{H}\mathbf{s} + \mathbf{w} \tag{2}$$

where $\mathbf{w}$ is a M-dimensional white Gaussion noise in the frequency domain, i.e., $\mathbf{w} = [\omega_1, \omega_2, \ldots, \omega_M]^T$.

For WiFi communication and wireless sensing, we can access the channel $\mathbf{H}$ through WiFi's Network Interface Controller (NIC) [47], which is also known as CSI. CSI describes the propagation process of the wireless signal and thus contains the geometric information of the propagation space. When signal $\mathbf{s}$ propagates through a multipath channel with P distinct paths and reaches the receiver, it can be characterized by CSI in the following aspects:

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Angle of Arrival θ_p: The phase shift of the signal arriving at each antenna is determined by the antenna spacing and the incoming angle. The phase shift of the mth receiving antenna relative to the first antenna, as a function of the pth path's AoA θ_p, is given by:

$$[\phi(\theta_p)]_m = e^{-j2\pi f_k \frac{(m-1)l\sin\theta_p}{c}} \tag{3}$$

where f_k is the subcarrier frequency of the signal, l is the interelement space of the antenna array, and c is the speed of light.

The vector of phase shifts for the whole array is:

$$\boldsymbol{\phi}(\theta_p) = \left[[\phi(\theta_p)]_1, [\phi(\theta_p)]_2, \ldots, [\phi(\theta_p)]_M\right]^T. \tag{4}$$

Time of Flight τ_p: The signal of each path will experience a different propagation time, which depends on the length of its path. In the frequency domain, the phase shift of kth subcarrier relative to the first subcarrier, as a function of the pth path's ToF τ_p, can be expressed as:

$$[\psi(\tau_p)]_k = e^{-j2\pi(k-1)\Delta f \tau_p} \tag{5}$$

where Δf is the frequency interval between consecutive subcarriers. The phase shift vector of the whole subcarrier dimension is:

$$\boldsymbol{\psi}(\tau_p) = \left[[\psi(\tau_p)]_1, [\psi(\tau_p)]_2, \ldots, [\psi(\tau_p)]_K\right]^T. \tag{6}$$

Complex attenuation γ_p: Each path p's signal will experience an attenuation γ_p.

Given the above parameters, we can represent the wireless channel $\mathbf{H}$ as follows:

$$\mathbf{H} = \sum_{p=1}^{P} \boldsymbol{\phi}(\theta_p)\boldsymbol{\psi}^T(\tau_p)\gamma_p. \tag{7}$$

B. Beamforming

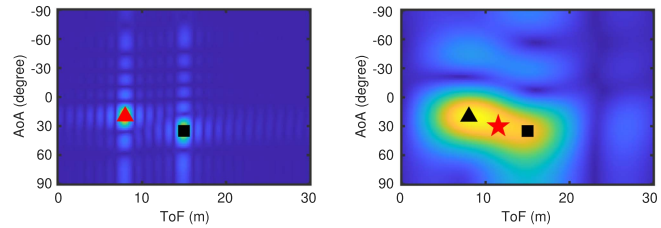
To extract the information needed for localization, CSI is usually transformed into a spectrum via beamforming [48], [49]. Recall that due to the propagation of the signal, the received signals introduce a phase shift in both the antenna and subcarrier dimensions, which can be characterized by CSI. By utilizing the phase offset information of CSI, we can extract location information. The measured CSI of the receiver is $\hat{\mathbf{H}}$, which is a matrix with dimension of $M * K$, and can be expressed as:

$$\hat{\mathbf{H}} = \begin{bmatrix} h_{1,1} & h_{1,2} & \ldots & h_{1,K} \\ h_{2,1} & h_{2,2} & \ldots & h_{2,K} \\ \vdots & \vdots & \ddots & \vdots \\ h_{M,1} & h_{M,2} & \ldots & h_{M,K} \end{bmatrix} \tag{8}$$

where $h_{m,k}$ represents the measured CSI corresponding to the mth antenna and kth subcarrier.

By compensating for the phase shift on different antennas and frequencies, the signals from AoA θ and ToF τ would superimpose coherently while the signals from other locations would be suppressed. The phase shift used to compensate for the phase can be expressed as:

$$\Phi_{m,k}(\theta,\tau) = \exp\left(-\mathrm{j}2\pi\left((\mathrm{k}-1)\Delta\mathrm{f}\tau + \mathrm{f_k}\frac{(\mathrm{m}-1)\mathrm{l}\sin\theta}{\mathrm{c}}\right)\right). \tag{9}$$



(a) Ideal spectrum (b) Resolution limitation

Fig. 2. Spectrum estimation problem simulation. (a) LOS path (red triangle) and multipath (black square) can be well distinguished with the setting of 8 antennas and the bandwidth of 250 MHz. (b) Under the configuration of 3 antennas and 40 MHz, the LOS path (black triangle) and multipath (black square) mix together resulting in an incorrect spectral peak (red star).

For a system with M antennas and K subcarriers, the phase shift vector can be written as:

$$\boldsymbol{\Phi}(\theta,\tau) = [\Phi_{1,1}(\theta,\tau), \ldots, \Phi_{M,1}(\theta,\tau), \ldots, \Phi_{M,K}(\theta,\tau)]^T. \tag{10}$$

By defining the spatial grid for candidate AoA–ToF values, the signals from these AoA–ToFs can be extracted by:

$$\mathbf{p} = \mathbf{A}^H \mathrm{vec}\left(\hat{\mathbf{H}}\right) \tag{11}$$

where vec$(\cdot)$ represents the vectorization operation, $(\cdot)^H$ is the Hermitian operator and $\mathbf{A}$ is defined as follows:

$$\mathbf{A} = [\boldsymbol{\Phi}(\theta_1,\tau_1), \boldsymbol{\Phi}(\theta_2,\tau_1), \ldots, \boldsymbol{\Phi}(\theta_{g_A},\tau_1), \ldots, \boldsymbol{\Phi}(\theta_{g_A},\tau_{g_T})] \tag{12}$$

where g_A and g_T denote the number of candidates AoA and ToF, respectively.

Finally, $\mathbf{p}$ can be reshaped into a matrix with dimensions of $g_A * g_T$, which is the AoA-ToF spectrum.

IV. Design

A. Problem Analysis

1) Signal Processing-Based Methods: Assuming that there is no complex multipath and the system's resolution is sufficient, then we can obtain an ideal spectrum as shown in Fig. 2(a). As we can see in Fig. 2(a), the Line-of-Sight (LOS) path (red triangle) and the multipath (black square) can be easily distinguished. However, the situation is not ideal in practice. If the direct path from the client to the AP is not separated by a certain distance from the reflected path in the environment, then the signal of these two paths will interfere with each other [17]. The interference effect can lead to the amplification or attenuation, and even disappearance, of the newly generated spectral peak's energy resulting from the superposition. One illustration of the interference effect, which leads to an enhancement in the energy of the newly generated spectral peak (red star) is illustrated in Fig. 2(b).

After obtaining the AoA and ToF value from the measured CSI of APs, the signal processing-based methods perform triangulation to locate the client. Here, the coordinates of the APs are known, and this process can be expressed as:

$$\mathbf{c} = f_{loc}(f_{spec}(\mathbf{x}), \boldsymbol{\kappa}) = f_{loc}(\boldsymbol{\theta}, \boldsymbol{\tau}, \boldsymbol{\kappa}) \tag{13}$$

where $\mathbf{c}$ represents the client's coordinates, $\mathbf{x}$ denotes the CSI of the APs, $\boldsymbol{\theta}$ represents the AoAs, $\boldsymbol{\tau}$ refers to the ToFs, $\boldsymbol{\kappa}$ represents

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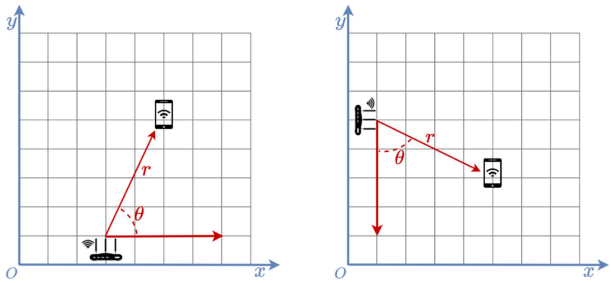
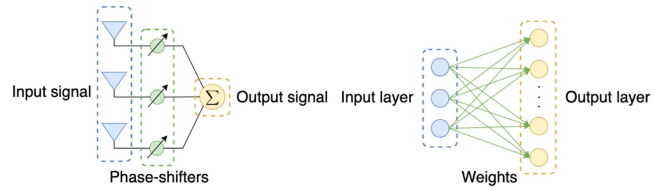


(a) Scenario 1 (b) Scenario 2

Fig. 3. Localization space of different scenarios, where Cartesian coordinate system and polar coordinate system are marked in blue and red, respectively. In this example, the Cartesian coordinates of AP and client are different while the APs-centered polar coordinate remains the same.



(a) Beamforming (b) Fully Connected Layer

Fig. 4. Beamforming versus fully connected layer.

the APs' coordinates, f_{loc} represents the localization function, and f_{spec} represents the function used for spatial spectrum estimation.

As we can see, the performance signal processing-based methods will degrade when the spatial resolution of the WiFi system is insufficient. This phenomenon will further deteriorate in complex environments with strong interference [17]. However, their mathematical models remain unchanged across different environments and thus can be deployed in any environment.

2) Deep Learning-Based Methods: Existing deep learning-based methods leverage the powerful fitting capability of neural networks to model the environment, and once the deep learning model is trained, it can provide location predictions based on the given input. Although these studies do not directly input the coordinates of APs into the network, they essentially use neural networks to fit the (13). As a result, the coordinates of APs become implicit parameters within the network function. Ultimately, the process of these methods can be described as follows:

$$\mathbf{c} = f_{\Theta}(\mathbf{x}, \boldsymbol{\kappa}) \tag{14}$$

where f_{Θ} represents a neural network with trainable parameters Θ.

Due to the complexity and diversity of indoor environments, the deployment of APs in different environments is usually different. Meanwhile, models trained within a single environment have already fitted the coordinates of APs, resulting in a strong correlation between the trained model and the training environment. This signifies that a well-trained model cannot be readily transferred to other environments.

3) Our Insight: Given the merits and limitations of both technical directions, we opt to integrate the cross-domain capability of signal processing and the fitting ability of deep learning techniques for reliable and high-precision localization.

The reason that signal processing-based methods can be directly deployed in different environments is that their mathematical models remain unchanged in different environments. This characteristic is ensured through the estimation of APs-centered polar coordinates, as Fig. 3 shows.

Based on this observation, we let the neural network estimate APs-centered polar coordinates. By this means, the network can focus solely on fitting polar coordinate location features weakly related to the environment and enhance the cross-domain capability of the model. After obtaining polar coordinates, we then perform localization based on the signal processing framework. This process can be described as follows:

$$\mathbf{c} = f_{loc}(f_{\Theta}(\mathbf{x}), \boldsymbol{\kappa}) = f_{loc}(\boldsymbol{\theta}, \boldsymbol{\tau}, \boldsymbol{\kappa}). \tag{15}$$

Based on this insight, we can leverage the powerful fitting capability of neural networks while avoiding overfitting the deep learning model to a single environment.

B. Unleashing the Potential of Neural Network by Beamforming Layer

When we use the neural network to process CSI, a straightforward method is to regard the number of antennas and subcarriers of CSI as the length and width of the image, and the real and imaginary parts of CSI as the different channels of the image [50], [51]. However, this method fails to leverage the knowledge of WiFi localization, which unnecessarily complicates the neural network used for localization [17]. In contrast, DLoc first utilizes 2D-FFT to transform CSI into the spatial spectrum and convert the amplitude of the spatial spectrum into a 2D Cartesian coordinates heatmap, which is then used as input to the neural network. This enables DLoc to incorporate the knowledge of WiFi-based localization, transform the localization problem into an image translation problem, and leverage various state-of-the-art models of deep learning to achieve high-precision localization. Nonetheless, DLoc solely relies on the amplitude of the spatial spectrum, which naturally entails information loss.

To this end, we carefully design a beamforming layer, which is a neural network-based approach to implement beamforming calculations. Our key insight is that the mathematical essence of beamforming and forward propagation of the fully connected layer is the same.

The core structure of the beamforming and fully connected layer is presented in Fig. 4. Specifically, the values of the input signal, output signal, and different phase-shifters of beamforming correspond to the values of input nodes, different output nodes, and different weights of the fully connected layer, respectively. In beamforming, the spatial spectrum is obtained by compensating the input signal's phase with different phase-shifters' values. Similarly, the processing procedure of beamforming can be viewed as a matrix multiplication operation between the input nodes of the layer and different weights of the fully connected layer. From this perspective, the forward propagation of a fully connected layer is the same as beamforming.

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In the fully connected layer, each input neuron is connected to every neuron in the next layer through weighted connections, which can be represented by a weight matrix. The output can be obtained by multiplying the input vector with the weight matrix and then adding the bias vector. This process can be expressed mathematically as shown in:

$$\mathbf{o} = \mathbf{W}\mathbf{i} + \mathbf{b} \tag{16}$$

where $\mathbf{o}$ represents the output vector, $\mathbf{W}$ is the weight matrix, $\mathbf{i}$ is the input vector, $\mathbf{b}$ is the bias vector.

Note that the main computational core of (11) and (16) is matrix multiplication. When the bias term of the fully connected layer is set to zero, we can use the fully connected layer to perform the computation described in (11), as shown in (17).

$$\mathbf{p} = \mathbf{W}\mathrm{vec}\left(\hat{\mathrm{H}}\right) \tag{17}$$

where $\mathbf{W}$ is the weight matrix used to replace $\mathbf{A}^H$ in (11), respectively. $\mathbf{p}$ represents the vectorized spatial spectrum, and $\hat{\mathbf{H}}$ refers to the measured CSI.

It is well known that the neural network can handle real values well. When we use the neural network to process complex-valued CSI, the structure of the neural network needs to be modified accordingly. One common representation of complex-valued numbers is the rectangular form $a + bi$, where a is the real part, b is the imaginary part, and i is the imaginary unit. When two complex-valued numbers are multiplied, we obtain the computation result shown in:

$$(a_1 + b_1 i)(a_2 + b_2 i) = (a_1 a_2 - b_1 b_2) + (a_1 b_2 + b_1 a_2)i \tag{18}$$

where $(a_1 + b_1 i)$ and $(a_2 + b_2 i)$ are two different complex-valued numbers.

When the computation involves complex-valued matrices, the corresponding equation becomes:

$$\left(W_1^a + W_1^b i\right)\left(W_2^a + W_2^b i\right) = \left(W_1^a W_2^a - W_1^b W_2^b\right) + \left(W_1^a W_2^b + W_1^b W_2^a\right) i \tag{19}$$

where $(W_1^a + W_1^b i)$ and $(W_2^a + W_2^b i)$ are two different complex matrices.

When we employ the neural network to perform the same operation, two bias-free fully connected layers can be used to separately represent the real and imaginary parts of the complex-valued matrix, and perform the computation described in (19). Additionally, to leverage the knowledge of beamforming, we initialize the weight matrices of the fully connected layers with the phase shift matrices of beamforming. The computational process achieved through the aforementioned method is referred to as the beamforming layer. By utilizing the beamforming layer, we can directly input the CSI without detailed feature extraction, benefit from advancements in the field of deep learning, and use the knowledge of beamforming. In this way, we can further unleash the potential of the neural network used for localization.

C. Deep Neural Network Design

Multi-Task Learning: Multi-task learning aims to improve learning efficiency and prediction accuracy of the neural network by learning multiple objectives from shared representations [52], and it has been widely applied in various fields such as computer vision [53] and natural language processing [54]. In this paper, we explore multi-task learning within the setting of deep learning-based localization.

Recalling recent advances in deep learning-based localization, a common approach is using the Cartesian coordinates to constrain the neural network to obtain features related to the absolute position. These methods achieve extremely high localization accuracy by allowing the network to learn features strongly correlated with the environment [17], [33]. However, once the position of the AP changes, the model obtained in this way cannot output the correct position corresponding to the features of input data. In contrast, we observe that APs-centered polar coordinates are independent of the position of APs. Once the neural network establishes a mapping relationship between data and APs-centered polar coordinates, the model can output the correct relative position based on the corresponding input data, regardless of changes in AP positions. This means that APs-centered polar coordinates can constrain the network to learn features related to relative positions. Therefore, the Cartesian coordinates and polar coordinates represent different information for the same location. To constrain the network to learn different features of both coordinates, we treat regressing the APs-centered polar coordinates and Cartesian coordinates of the client as two branches of the network. By leveraging the idea of multi-task learning, we can get a model that achieves both high-precision localization and excellent cross-domain capability.

Network Architecture: We model the location parameter estimation task as multi-label regression. The input of our model is the real and imaginary parts of CSI and the labels are polar coordinates and Cartesian coordinates. Given the input CSI, we first use the beamforming layer to get the spatial spectrum. Then, we consider the spatial spectrum as an image and take its real part, imaginary part, and amplitude as the three channels of the image. We choose to use the ResNet [55] architecture, which has been widely validated and has become a cornerstone of deep learning, as the network for feature extraction. Specifically, we employ 1 convolutional layer followed by 4 ResNet BasicBlocks for feature extraction.Finally, we use several fully connected layers with two branches to regress the mean and variance of location parameters.

Through such a data-driven approach, we can utilize the neural network to create an implicit representation of the environment and incorporate the ground truth data observed from the real world to extract the impact of the environment on CSI. By employing the trained neural network, we can use the phase and amplitude of CSI to obtain high-precision AoA and range.

Confidence Estimation: During testing, the confidence of different APs in providing accurate location information is different due to differences in received signal-to-noise ratios and whether the AP is obstructed. However, the confidence of different APs in providing location information is not equal [56], [57], [58], [59]. Existing signal processing-based and deep learning-based works have employed various approaches to quantify the confidence of different APs [33], [56], [57], [58], [59]. In this paper, we adopt a probabilistic approach [33] for multi-AP localization, following the principles of probability theory. For a regression task, we make the last layer of the neural network output two values, one representing the mean and the other representing the variance [60]. By treating the input data as a sample from

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a Gaussian distribution with predicted mean and variance, the neural network is trained to minimize the negative log-likelihood (NLL) criterion of (20). Once trained, the neural network can output both the predicted mean and variance of J APs.

$$NLL(y_j, \mathbf{x}) = \frac{\log\sigma_j^2(\mathbf{x})}{2} + \frac{(y_j - \mu_j(\mathbf{x}))^2}{2\sigma_j^2(\mathbf{x})} + \text{constant} \quad (20)$$

where y_j denotes the real-value label of jth AP, the constant is a bias parameter, $\mu_j(\mathbf{x})$ and $\sigma^2(\mathbf{x})$ represent the predicted mean and variance of the neural network for jth AP, respectively.

Intuitively, the larger the variance, the less reliable the corresponding location-related information. After obtaining the variance of each AP, we can calculate the confidence of jth AP using the (21).

$$\text{Confidence}_j = \frac{1}{\sigma_j^2}. \quad (21)$$

Loss Function: The neural network is trained end-to-end and the final loss term $\mathcal{L}$ is a combination of the polar coordinates loss and Cartesian coordinates loss:

$$\mathcal{L} = NLL(\theta_j, \mathbf{x}) + \lambda_1 NLL(d_j, \mathbf{x}) + \lambda_2 \frac{(NLL(c_x, \mathbf{x}) + NLL(c_y, \mathbf{x}))}{2} \quad (22)$$

where θ_j and d_j represent the ground truth of AoA and range of Jth AP, respectively. c_x and c_y are two-dimensional Cartesian coordinates. λ_1 and λ_2 are weight factors and can be used to adjust the proportion of loss of different tasks in the final loss.

D. Localization

As mentioned earlier, polar coordinates and Cartesian coordinates are interchangeable. To unify the estimated coordinates of J APs, we convert the polar coordinates of each AP to Cartesian coordinates based on the geometric model of localization:

$$\begin{cases} \overline{c}^j_{x,\text{indirect}} = c^j_x + \overline{d}_j \sin\left(\overline{\theta}_j + \frac{\pi}{2}\text{sgn}(v^j_x)\right) \\ \overline{c}^j_{y,\text{indirect}} = c^j_y + \overline{d}_j \cos\left(\overline{\theta}_j + \frac{\pi}{2}\text{sgn}(v^j_y)\right) \end{cases} \quad (23)$$

where $\overline{c}^j_{x,\text{indirect}}$ and $\overline{c}^j_{y,\text{indirect}}$ are the mean value of abscissa and ordinate converted from polar coordinates of AP_j, respectively. c^j_x and c^j_y represent AP_j's abscissa and ordinate. $\overline{d}_j$ and $\overline{\theta}_j$ are the estimated mean value of AP_j's distance and angle, respectively. To determine the unique transformation function for each AP, v^j_x and v^j_y are used to determine the values of the sign function $\text{sgn}(\cdot)$.

Then, we apply error propagation theory [61] to convert the variance of polar coordinates of AP_j into the variance of Cartesian coordinates by:

$$\begin{aligned} \sigma^2_{c^j_{x,\text{indirect}}} &= \left(\frac{\partial \overline{c}^j_{x,\text{indirect}}}{\partial \overline{d}_j}\sigma_{d_j}\right)^2 + \left(\frac{\partial \overline{c}^j_{x,\text{indirect}}}{\partial \overline{\theta}_j}\sigma_{\theta_j}\right)^2 \\ \sigma^2_{c^j_{y,\text{indirect}}} &= \left(\frac{\partial \overline{c}^j_{y,\text{indirect}}}{\partial \overline{d}_j}\sigma_{d_j}\right)^2 + \left(\frac{\partial \overline{c}^j_{y,\text{indirect}}}{\partial \overline{\theta}_j}\sigma_{\theta_j}\right)^2 \end{aligned} \quad (24)$$

where ∂ represents the partial derivative, and σ denotes the standard deviation. Through this formula, we get the variance of polar coordinates in Cartesian coordinates $\sigma^2_{c^j_{x,\text{indirect}}}$ and $\sigma^2_{c^j_{y,\text{indirect}}}$.

Finally, we perform the probabilistic fusion with J APs using Cartesian coordinates obtained by direct regression and indirect transformation with their corresponding variances to get the Cartesian coordinates of the client, as shown in:

$$\begin{cases} c_{x,\text{client}} = \dfrac{\sum_{j=1}^{J}\left(\sigma^2_{c^j_{x,\text{direct}}}\right)^{-1}\overline{c}^j_{x,\text{direct}} + \sum_{j=1}^{J}\left(\sigma^2_{c^j_{x,\text{indirect}}}\right)^{-1}\overline{c}^j_{x,\text{indirect}}}{\sum_{j=1}^{J}\left(\sigma^2_{c^j_{x,\text{direct}}}\right)^{-1} + \sum_{j=1}^{J}\left(\sigma^2_{c^j_{x,\text{indirect}}}\right)^{-1}} \\ c_{y,\text{client}} = \dfrac{\sum_{j=1}^{J}\left(\sigma^2_{c^j_{y,\text{direct}}}\right)^{-1}\overline{c}^j_{y,\text{direct}} + \sum_{j=1}^{J}\left(\sigma^2_{c^j_{y,\text{indirect}}}\right)^{-1}\overline{c}^j_{y,\text{indirect}}}{\sum_{j=1}^{J}\left(\sigma^2_{c^j_{y,\text{direct}}}\right)^{-1} + \sum_{j=1}^{J}\left(\sigma^2_{c^j_{y,\text{indirect}}}\right)^{-1}} \end{cases} \quad (25)$$

The framework of the proposed method is shown in Fig. 5.

V. Implementation

A. Experiment Setup

1) Dataset: To evaluate the performance of our method, we conduct extensive experiments on three different datasets. Among them, the DLoc dataset[1] introduces simultaneous localization and mapping (SLAM) into deep learning-based localization to achieve automatic and high-precision data acquisition. Compared with previous single-point data acquisition, it can evaluate the performance of localization methods more comprehensively. The DLoc dataset contains two scenarios, one is the simple LOS scenario and the other is the high-multipath and NLOS scenario. To evaluate the generalization of the model over time in a single scenario, there are some different setups for each scenario. It is worth noting that the DLoc dataset has two limitations: 1) the moving speed of the robot is 15 cm/s which is too slow 2) there is no interference from the human body's movement. Therefore, the DLoc dataset is not close to the real world.

Taking this into consideration, we construct a human-held device dataset based on Ultra Wide Band (UWB) and WiFi, with 2D Cartesian coordinates ground truth provided by UWB [62]. For more details, please refer to the link.[2] We have also developed a camera-WiFi data collection system, where the ground truth of the data is obtained using computer vision algorithms based on the camera data [3]. For the settings of this data collection system, there are 12 cameras around the room and 4 WiFi APs with 1.2 m height placed near the horizontal center of each wall in the room. Cameras are calibrated before data collection and synchronized with APs through the NTP protocol. Each AP is a mini-computer equipped with a commodity Intel 5300 WiFi NIC, which has a ULA of three antennas, and each antenna is placed with a spacing of 2.6 cm (nearly half the wavelength of the 5.3 GHz band). The mini-computer measures the CSI of each packet from the physical layer of NIC using the CSI tool [47] with 40 MHz bandwidth on channel 60. We use another mini-computer with the same configuration as the client whose location is of interest to keep transmitting packets. The WiFi NICs of all APs operate in monitor mode and keep listening to the 5 GHz wireless channel. When constructing these two datasets, how volunteers held the transmitter is depicted in Fig. 7.

[1] https://wcsng.ucsd.edu/dloc/
[2] https://github.com/H-WILD/human_held_device_wifi_indoor_localization_dataset
[3] https://anonymous.4open.science/r/Camera-WiFi_localization_dataset-8E5F

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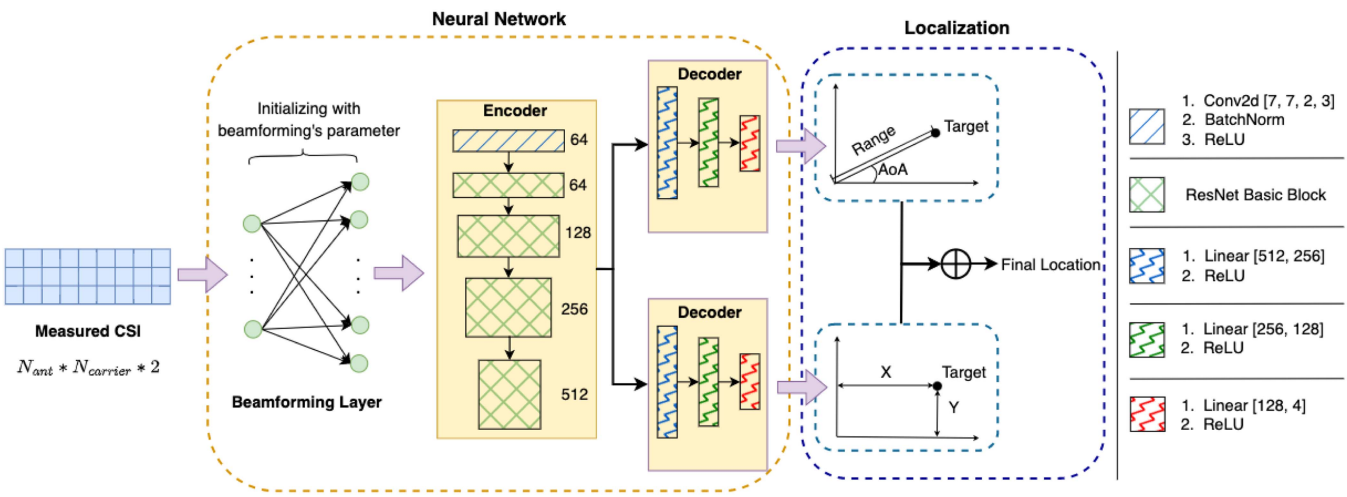
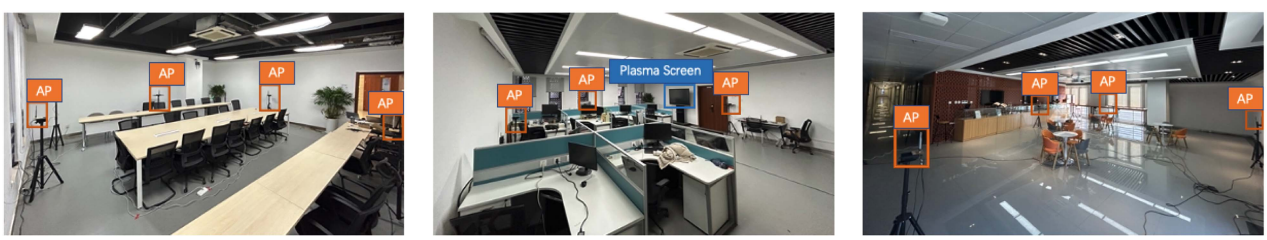


Fig. 5. Framework of the proposed methods. First, the measured CSI is fed into the neural network and processed by the beamforming layer, followed by feature extraction through the encoder. Next, the network utilizes the decoder to separately regress polar coordinates and Cartesian coordinates. Finally, the localization stage integrates multiple sources of information for localization. For each Conv2d, the four values shown are [$kernel\text{-}height$, $kernel\text{-}width$, $stride$, $padding$] and the black number in the encoder block indicates the number of kernels. While for each linear layer, the two key values are [$input\text{-}dimension$, $output\text{-}dimension$].



(a) Conference (b) Office (c) Lounge

Fig. 6. UWB-WiFi dataset different scenes' layout. The APs are marked in orange, and the reflectors are marked in blue.

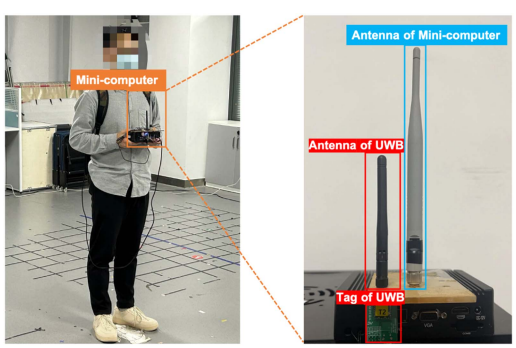


Fig. 7. Hand-held diagram of the transmitter.

2) Training and Testing Details: We implement our model using PyTorch [63] framework and set the activation function as ReLU and the dropout rate as 0.2. The training epoch of the model is 25 and the batch size is 256. The model is trained with an Adam optimizer with a learning rate of 0.01 and the learning rate is adjusted by exponential attenuation with a decay factor of 0.99. The weight factors λ_1 and λ_2 are both set to 2. The ratio of the training set to the test set is 8:2 and we choose the latest model as the final model.

3) Phase Calibration: For the phase calibration of APs as mentioned in SpotFi [13] and BreathTrack [64], we measure all the possible phase differences across antennas of each AP in advance [65]. When the device restarts, we will carry the transmitter to move in front of the AP to collect data. After that, we use the phase difference obtained in advance to calibrate the restarted AP and run the beamforming algorithm to estimate AoAs. We select the phase difference with minimum AoA estimation error as the phase difference for this startup of AP. This is a more efficient method than manually connecting the coaxial cable calibration.

B. Baselines

We benchmark the performance of our method against the following state-of-the-art indoor localization methods.

Co-Loc [16] introduces a non-parametric sparse recovery metric to quantify the accuracy of AoA and further improves the accuracy of AoA and localization by observing that the AoA between the user client and multiple APs intersect at one point.

DLoc [17] defines deep learning-based localization as image translation and borrows existing neural network architectures to achieve high localization accuracy.

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TABLE I
SCENARIO CLASSIFICATION

Category	Scenario	Space Size	Description	Pedestrian Occlusion	Training / Test Data
Simple	Atkison Hall	8m × 5m	A small hall for pedestrians.	No	19798 / 4950 samples.
	Conference	8m × 8m	A small seminar room for conference.	Yes	21547 / 5387 samples.
	Lounge	11m × 14m	A large open space for relaxation.	Yes	34043 / 8511 samples.
Complex	Jacobs Hall	18m × 8m	A large hall for pedestrians with several distinct large reflectors.	No	39926 / 9982 samples.
	Office	9m × 11m	A room for daily work with one reflector and several cluttered workstations.	Yes	20084 / 5022 samples.

SpotFi [13] utilizes spatial smoothing MUSIC algorithm to jointly estimate AoA and ToF information from CSI and combines AoA from several APs to determine the client location.

After we evaluate the overall performance of our method, we delve into the performance of individual parameter estimation accuracy (i.e., AoA and range). In particular, we compare the AoA estimation accuracy with the estimated results of beamforming, SpotFi, and Co-Loc. Then, we compare range estimation accuracy with the estimated results of CUPID [66].

The data format of the DLoc dataset and our dataset are different, so some of the methods are modified, and we make sure to replicate their works as best we can.

VI. EVALUATION

For a comprehensive performance comparison between our methods and state-of-the-art localization methods, we divide the test scenarios into simple scenarios and complex scenarios. In the simple scenarios, the Atkison Hall of DLoc dataset and UWB-WiFi dataset's Conference and Lounge are selected. There are no large-scale reflectors in these scenarios, which corresponds to good conditions for testing localization methods normally. For complex scenarios, we select the Jacobs Hall of DLoc dataset and the Office of UWB-WiFi dataset. There are several reflectors that can cause complex multipath propagation in these scenarios. This corresponds to a more challenging and practical deployment for a WiFi-based localization system. The detailed description of each scenario can be found in Table I.

Testing the generalization performance of deep learning-based methods is of utmost importance because their ability to effectively generalize to unseen data is a critical factor in their practical deployment. Evaluating the generalization performance allows us to assess how well these models can handle variations, noise, and different scenarios that may be encountered in real-world applications. In this paper, we test their generalization across domains and layouts.

And then, we delve into our method through an ablation study to test the effectiveness of our method. Finally, we perform some experiments with other considerations such as computational consumption.

A. Simple Scenarios

1) Localization Accuracy: To validate the correctness of our replication of existing work and compare the fundamental performance of various methods, we choose simple scenarios of the datasets for this experiment. For Atkison Hall, 3 APs are deployed and the environment is very simple, with LOS condition. For the Conference, it deploys 4 APs and the human body's walking route is very simple. For Lounge, its available space for the human body's movement is further enlarged with 4 APs deployed and it includes multiple tables surrounded by chairs. The localization results of these scenarios are shown in Fig. 8.

Atkison Hall: In Atkison Hall, our method's median error is 0.44 m and the 90th percentile error is 0.80 m. DLoc's median error is 0.41 m and the 90th percentile error is 0.85 m. Our method's median error is slightly worse than DLoc and the 90th percentile error is slightly better than DLoc. On the other hand, SpotFi and Co-Loc have a median error of 0.99 m and 0.88 m and a 90th percentile error of 2.02 m and 1.71 m. We can observe that deep learning-based methods significantly outperform signal processing-based methods, and the signal processing-based methods do not achieve the same level of localization accuracy but instead barely reach decimeter-level precision. First, this is due to the strong fitting capability of deep learning-based methods to the environment. Second, the data collected by the random moving robot and the human body is not biased and can be from any point in space. Such data may fall into boundary angles of the array, leading to large AoA estimation errors and decreased localization performance for methods that do not consider this situation [57].

Conference: When the testing scenario changes to Conference, our method maintains the same performance as tested in Atkison Hall. Specifically, the median error of our method is 0.38 m, and the 90th percentile error is 0.81. As for DLoc, its median error remains the same which is 0.41 m and the 90th percentile error grows to 1.17 m. The main difference between the two testing scenarios is whether the device is held by the human body or not. DLoc combines the information from different APs and feeds it into the network for training. It is worth noting that the CSI of any AP is affected by human movements, which leads to differences between the combined features during the testing phase and what the network has learned. As a result, it causes a decrease in localization performance. Our method decouples the AP information at the input of the network and only performs localization based on the individual localization results of different APs during the localization phase. This reduces the mutual impact between different AP information and thus improves localization accuracy. Due to the limited spatial resolution of WiFi devices and the lack of modeling for the human body's disturbance, SpotFi and Co-Loc suffer more severe performance degradation, with median errors of 1.58 m and 1.49 m and 90th percentile errors of 3.60 m and 3.89 m, respectively. The

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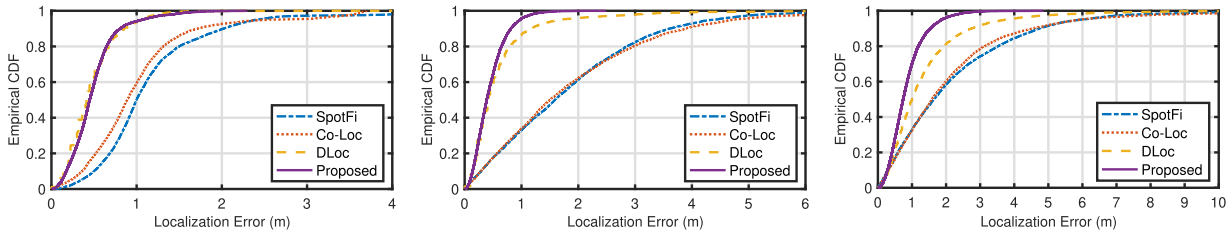


(a) Atkison Hall (b) Conference (c) Lounge

Fig. 8. Localization performance in simple scenarios.

localization accuracy of Co-Loc becomes nearly the same as or even worse than SpotFi, indicating that the method proposed in Co-Loc to improve localization performance based on the observations of multiple APs' AoAs will intersect at one point is ineffective in the presence of human body's disturbance.

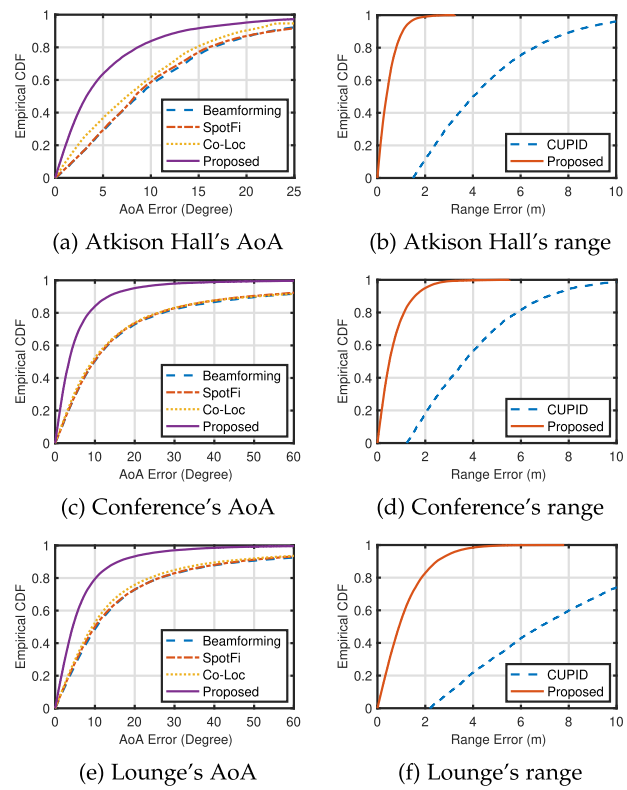
Lounge: For the Lounge with a larger spatial area, our method's median error and 90th percentile error increase to 0.73 m and 1.53 m respectively. DLoc's median error increases to 1m while and 90th percentile error decreases to 2.74 m. SpotFi and Co-Loc's performance further decreases with a median error of 1.63 m and 1.58 m and a 90th percentile error of 4.73 m and 4.54 m, respectively. The expanded open area of the Lounge introduces greater complexity to the random walking paths of individuals. This further amplifies the impact of the human body's movement on the CSI, increasing the complexity of the neural network's fitting function. As a result, the performance of deep learning-based methods deteriorates. Meanwhile, the localization errors of signal processing-based methods continue to increase, further indicating the significant impact of the human body's disturbance on the performance of these methods.

2) Parameter Estimation Accuracy: These scenarios' parameter estimation results are shown in Fig. 9. It can be seen from the results that our parameter estimation performance is significantly better than other methods, which serves as the foundation for our high-precision localization.

AoA: For AoA estimation, our method's median error maintains at 3–4 degrees and the 90th percentile error maintains at 10–16 degrees across different scenarios. Beamforming, SpotFi, and Co-Loc exhibit similar levels of performance, with median errors ranging from 5 to 10 degrees, which is not significantly different from our method. However, they all have relatively large 90th percentile errors, ranging from 20 to 51 degrees. For joint multi-AP-based localization using AoA, a single large error in one AoA measurement can lead to significant errors in the final localization result. This is precisely why these AoA-based methods perform poorly in complex scenarios.

Range: For range estimation, our method's median error maintains at 0.38–0.90 m and the 90th percentile error maintains at 1.05–2.48 m across different scenarios. As for CUPID, as shown in Fig. 11, its performance is poor, rendering it inadequate for practical indoor location-based services.

Our approach achieves high-precision parameter estimation through unleashing the potential of the neural network by beamforming layer. This consistently leads to superior localization performance compared to other methods, regardless of the scenarios.

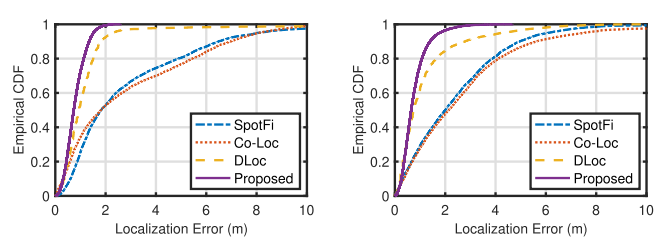


(a) Atkison Hall's AoA (b) Atkison Hall's range

(c) Conference's AoA (d) Conference's range

(e) Lounge's AoA (f) Lounge's range

Fig. 9. Simple scenarios' empirical CDF for AoA and range error.



(a) Jacobs Hall (b) Office

Fig. 10. Localization performance in complex scenarios.

B. Complex Scenarios

1) Localization Accuracy: The real environment is not always as ideal as the simple scenarios, there may be a variety of complex multipath propagation, huge reflecting surfaces, and other factors affecting the quality of data. To this end, we design complex scenarios to test the performance of multiple methods

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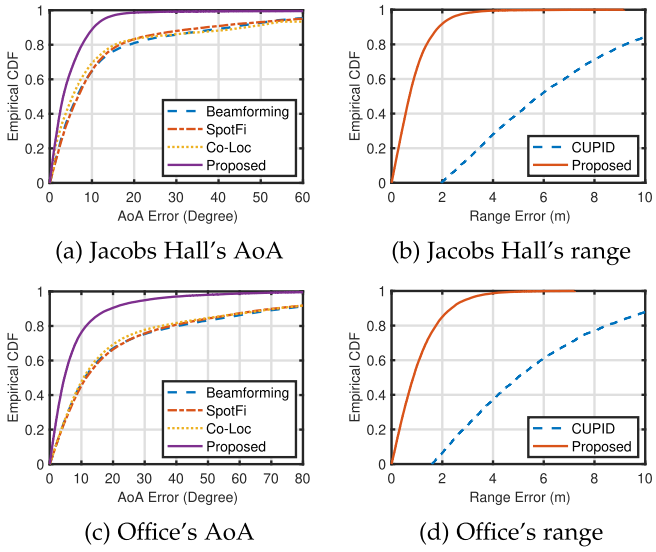

Fig. 11. Complex scenarios' empirical CDF for AoA and range error.

under challenging conditions. These scenarios' localization results are shown in Fig. 10.

Jacobs Hall: For Jacobs Hall, there are two plasma screens and one AP blocked by the wall. Our method's median error is 0.73 m, while for DLoc, SpotFi, and Co-Loc, it is 0.94 m, 1.82 m, and 1.81 m, respectively. Moreover, the 90th percentile error of our method is 1.34 m, whereas the 90th percentile error of DLoc, SpotFi, and Co-Loc are higher at 1.81 m, 6.48 m, and 6.96 m. In this scenario, the performance of DLoc is slightly inferior to our method. Meanwhile, the performance of signal processing-based methods performs poorly, indicating that they are unable to handle complex scenarios with numerous reflective surfaces.

Office: Office introduces the human body's disturbance, reducing one reflective surface and contains several cluttered workstations. Our method's median error is 0.64 m and the 90th percentile error of 1.40 m. DLoc's median error reduces to 0.76 m, but the 90th percentile error increases to 2.79 m than in Jacobs Hall. This further indicates that the human body's disturbance will increase the tail error of DLoc. As for SpotFi and Co-Loc, their median error is 1.99 m and 2.14 m and the 90th percentile error is 4.95 m and 5.63 m, respectively. Compared to the testing results in Jacobs Hall, their tail error has decreased. We attribute this to the reduction in the number of reflections, which makes it easier for traditional signal processing-based methods to obtain the LOS path's AoA from CSI, thereby mitigating the degradation in localization performance.

2) Parameter Estimation Accuracy: AoA: For AoA estimation, our method obtains a median error of 3.2 degrees and a 90th percentile error of 10.4 degrees in Jacobs Hall. As for Co-Loc in Jacobs Hall, its median error is 5.4 degrees and 90th percentile error of 46.2 degrees. SpotFi and beamforming have nearly the same performance with the median error of 6.7 and 6.5 degrees and the 90th percentile error of 36.7 degrees and 42.6 degrees, respectively. Our method's performance slightly deteriorates when the testing scenario changes to Office with a median error of 4.6 degrees and a 90th percentile error of 19.3 degrees. However, the performance of Co-Loc, SpotFi, and beamforming has been significantly decreased. Specifically, Co-Loc, SpotFi, and beamforming's median error increases by 5.3 degrees, 4.8 degrees, and 4.6 degrees, and 90th percentile error increases by 25.1 degrees, 33.8 degrees, and 31.1 degrees.

Range: For range estimation, we get a median error of 0.70 m and a 90th percentile error of 1.86 m in Jacobs Hall for our method. While in Office, our method's median error is 0.86 m and the 90th percentile error is 2.36 m.

C. Generalization

We propose to test the generalization performance of deep learning-based localization methods from two distinct perspectives: domain generalization and layout generalization. These perspectives encompass different dimensions of generalization, ensuring a comprehensive evaluation of the method's ability to perform accurately and consistently across various domains and layouts. By examining the deep learning-based method's performance in these diverse scenarios, we can gain insights into its adaptability, robustness, and suitability for real-world applications.

1) Domain Generalization: Training a deep learning model typically requires a large amount of data and significant computational resources. If a deep learning model trained in one domain can directly generalize to other domains, it would greatly save human and computational resources. To test this capability of the deep learning-based methods, we train the model in a single scenario and test it in other scenarios to evaluate its cross-domain performance.

As mentioned earlier, the Cartesian coordinates output by the neural network are highly dependent on the training environment. Therefore, we only use the polar coordinates output by our neural network to test cross-domain performance. The results are shown in Table III. Regardless of the training and testing scenarios, our method demonstrates basic localization capabilities for decimeter-level localization, although there is a certain decrease in performance compared to the source scenario. On the other hand, DLoc exhibits a significant increase in localization error. In the best case, the median error and 90%−tail error of our method increases by 69% and 73%, respectively. In contrast, the median and 90%−tail error of DLoc increases by 195% and 150%, respectively. The root cause of this disparity lies in the fact that DLoc takes as input and output position heatmaps in the Cartesian coordinates. And these heatmaps embed the location of the APs, which are likely to vary in different environments. As a result, position heatmaps become features strongly correlated with the source scenario. By incorporating angular and distance information in the polar coordinates as outputs, we enable the network to learn domain-generalizable features.

2) Layout Generalization: Once deep learning-based localization methods are deployed, daily activities can lead to changes in the environment, resulting in inconsistencies between the deployment of the scene and the training scene. Since we cannot update the model in real-time, testing the generalization ability of deep learning-based localization methods to different scene deployments within the same scenario is crucial. We refer to this generalization ability as layout generalization.

To evaluate this ability, we set up 7 different layouts within one scenario and collect corresponding data for training and testing

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TABLE II
LAYOUT GENERALZIATION TESTING

Label	Setting	Median Error(m)		90%−tail Error(m)	
		Proposed	DLoc	Proposed	DLoc
1	-	**0.69**	0.72	**1.18**	1.68
2	Chair	**0.78**	1.06	**1.31**	2.09
3	Chair, Bed	**0.75**	0.94	**1.23**	2.01
4	Chair, Plasma screen	**1.03**	1.33	**1.92**	3.98
5	Chair, Plasma screen, White board	**1.06**	1.39	**2.03**	4.28
6	Chair, Plasma screen, White board, Radar	**1.19**	1.54	**2.47**	3.84
7	Chair, Plasma screen, White board, Radar, Bed, Vector Network Analyzer	**0.94**	1.22	**1.63**	3.42

TABLE III
DOMAIN GENERALZIATION TESTING

Train Scene	Test Scene	Median Error(m)		90%−tail Error(m)	
		Proposed	DLoc	Proposed	DLoc
Conference	Conference	**0.36**	0.39	**0.81**	1.19
	Office	**1.55**	4.18	**2.79**	7.60
	Lounge	**1.81**	3.20	**3.62**	7.00
Office	Office	**0.65**	0.79	**1.44**	2.81
	Conference	**0.90**	3.04	**1.85**	5.48
	Lounge	**1.47**	4.02	**3.15**	8.31
Lounge	Lounge	**0.75**	1.03	**1.55**	2.76
	Conference	**1.17**	2.82	**2.08**	6.02
	Office	**1.27**	3.04	**2.69**	6.92

using the camera-WiFi system described earlier. Among them, the first layout is empty, while the second to the seventh layouts sequentially introduce different furniture and experimental equipment. The training dataset consists of 21 600 samples, and the dataset of each test scenario consists of 5,400 samples. The final testing results are shown in Table II.

In terms of the test results, our method consistently outperforms DLoc. There are two notable points to mention. First, as the environment becomes more complex, the localization errors of both methods increase, but they are not strictly positively correlated. Second, in the test results of scene 4, the introduction of a large reflective surface, namely a plasma screen, leads to an increase in both median error and 90th percentile error for both methods, demonstrating the significant impact of reflective surfaces on deep learning-based methods.

D. Ablation Study

We conduct a series of controlled experiments to evaluate the impact of different components or factors on the performance of our proposed method. For ablation experiments, we perform comprehensive testing using data from both simple and complex scenarios.

1) Ablation of Beamforming Layer: To confirm the effectiveness of the beamforming layer, we conducted two distinct experiments. One involves removing the beamforming layer from the network and directly inputting the CSI complex spectrum obtained through beamforming. The other experiment entails substituting the beamforming layer with the convolutional layer proposed in [50]. The results are shown in Table IV.

Compared to the convolutional layer, the beamforming layer reduces the median error by 9% to 26% and decreases the 90th percentile error by 10% to 28%. Compared to directly inputting the complex spectrum, the beamforming layer decreases the median error by 6%–39% and decreases the 90th percentile error by 8%–44%. It can be seen from the results that the beamforming layer effectively improves the localization accuracy.

2) Ablation of Beamforming Initialization: In the previous chapters, we initialize the weight matrix of the beamforming layer with the phase shift matrix of beamforming. To validate the effectiveness of the knowledge of beamforming, we compare the results of randomly initializing the beamforming layer with the results of initializing it with beamforming's phase shift matrix. From the results in V, it is evident that the beamforming initialization improves the performance of the model. Specifically, the beamforming initialization reduces the median error by 9% to 14% and decreases the 90th percentile error by 5% to 26%.

3) Ablation of Multi-Task Learning: To test the effectiveness of multi-task learning, we made modifications to the proposed network. The modified network consists of two variants: one variant regresses the polar coordinates, while the other variant directly regresses the Cartesian coordinates. These two networks are referred to as the AoA-Range branch network and the X-Y branch network, respectively. The test results are shown in Table VI.

One intuitive observation from the results is that the performance of the network using multi-task learning is not always the best. In some cases, the network directly regressing Cartesian coordinates performs better. However, this is because the neural network overfits the environment, which also makes the neural network lack domain generalization. Our goal is to achieve both high localization accuracy and domain generalization through multi-task learning, and it effectively combines these two aspects. Compared to the network that directly regresses Cartesian coordinates tightly coupled with certain scenarios, the accuracy of the multi-task learning network with cross-domain capability exhibits a slight decrease in some scenarios, which is acceptable.

E. Other Considerations

1) Transfer the Model to New Domain: In Section VI-C, we have already demonstrated the domain generalization ability of our method. Here, we will show that by fine-tuning the model with only a small portion of data from the new environment, our model can further improve the localization accuracy.

From the results in Fig. 12, it can be observed that with the increase in the amount of training data, the median error and 90th percentile error of the fine-tuned models show a decreasing trend. When the training data amounts to 10%, it leads to a reduction of 1%–26% and 9%–25% in the median error and 90th percentile error, respectively, for most scenarios. While there are cases where the errors increase by 3%–12%. When the training data amounts to 20%, the median error

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TABLE IV
ABLATION OF BEAMFORMING LAYER

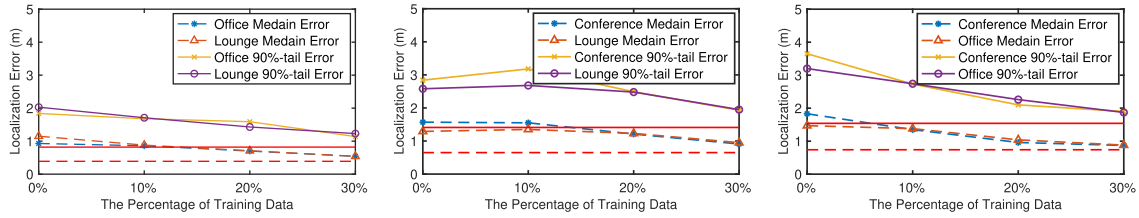
Train&Test Scene	With Beamforming Layer		With Convolutional Layer		Without Beamforming Layer	
	Median Error(m)	90%−tail Error(m)	Median Error(m)	90%−tail Error(m)	Median Error(m)	90%−tail Error(m)
Atkison Hall	**0.42**	**0.76**	0.46	0.85	0.45	0.83
Jacobs Hall	**0.73**	**1.35**	0.85	1.65	0.78	1.52
Conference	**0.37**	**0.79**	0.47	0.95	0.54	1.10
Office	**0.62**	**1.29**	0.84	1.80	1.02	2.32
Lounge	**0.71**	**1.49**	0.81	1.74	1.09	2.30

TABLE V
ABLATION OF BEAMFORMING INITIALIZATION

Train&Test Scene	With Beamforming Initialization		Without Beamforming Initialization	
	Median Error(m)	90%−tail Error(m)	Median Error(m)	90%−tail Error(m)
Atkison Hall	**0.42**	**0.76**	0.46	0.84
Jacobs Hall	**0.73**	**1.35**	0.85	1.82
Conference	**0.37**	**0.79**	0.40	0.83
Office	**0.62**	**1.29**	0.71	1.42
Lounge	**0.71**	**1.49**	0.79	1.68

TABLE VI
ABLATION OF MULTI-TASK LEARNING

Train&Test Scene	With Multi-task Learning		With AoA-Range Constraints		With X-Y Constraints	
	Median Error(m)	90%−tail Error(m)	Median Error(m)	90%−tail Error(m)	Median Error(m)	90%−tail Error(m)
Atkison Hall	0.44	0.80	0.44	0.87	**0.40**	**0.70**
Jacobs Hall	**0.73**	**1.34**	0.76	1.50	0.84	1.52
Conference	0.38	0.81	0.38	0.82	**0.36**	**0.77**
Office	**0.64**	**1.40**	0.70	1.57	0.67	1.48
Lounge	0.73	1.53	0.78	1.67	**0.70**	**1.50**
Mean of Scenarios	**0.58**	**1.18**	0.61	1.29	0.59	1.19



(a) Conference (b) Office (c) Lounge

Fig. 12. Impact of the percentage of training data from the target domain on localization. The median and 90th percentile errors of the model trained solely on the target domain data are indicated by a red dashed horizontal line and a solid horizontal line, respectively. Dashed and solid lines with markers represent the median and 90th percentile errors of the target domain's fine-tuned models.

decreases by 5%–47%, and the 90th percentile error decreases by 4%–42%. With a training data amount of 30%, the median error and 90th percentile error remain within 1 m and 2 m, respectively. Nonetheless, in comparison to the models trained solely in the target domain, the median error and 90th percentile error still exhibit differences of 0.15 m–0.3 m and 0.36–0.54 m, respectively. This is just a preliminary validation, and we believe that there are more effective methods to transfer models with domain-invariant features to new environments.

2) Impact of AP Number: In the actual environment, it is not always possible to use multiple APs' information, so it is meaningful to study how the number of APs affects the localization accuracy of our method. We conduct experiments based on the data collected by the camera-WiFi system, and the results are given in Table VII.

TABLE VII
DIFFERENT AP NUMBER'S TEST RESULT

AP Number	Median Error(m)	90%−tail Error(m)
1	2.81 ± 0.32	4.79 ± 0.38
2	1.30 ± 0.21	2.11 ± 0.29
3	0.88 ± 0.09	1.63 ± 0.16
4	0.68	1.25

In terms of the results, we can observe that the localization accuracy improves with the increase in the number of APs. This is because the increase in the number of APs makes more information available and reduces the negative impact of data noise on localization results.

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TABLE VIII
COMPARISON OF THE MODEL SIZE AND INFERENCE TIME

Model	Model Size(MB)	Inference Time(ms)
Proposed	**5.31**	**0.11**
DLoc	16.48	3.38

TABLE IX
SCENARIO DESCRIPTION

Scenario	Space Size	Pedestrian Occlusion	Training / Test Data
Office	9m × 11m	Yes	20084 / 5022 samples.
Hall	**36m × 12m**	Yes	15473 / 3869 samples.

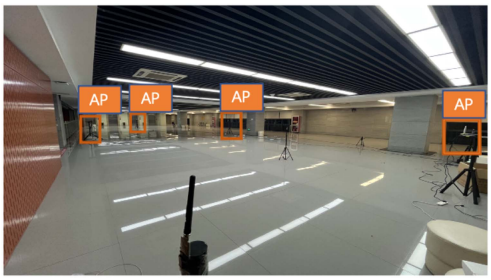
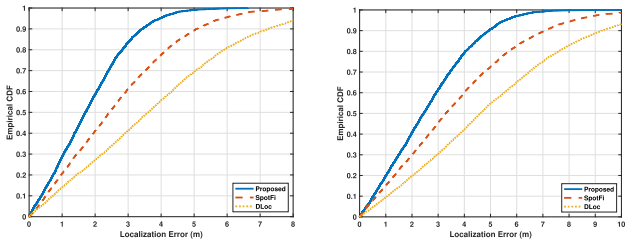


Fig. 13. The layout of the Hall scenario.



(a) Office (b) Hall

Fig. 14. Cross-domain evaluation of large-scale experiments. (a) Training in the Hall, while testing in the Office. (b) Training in the Office, while testing in the Hall.

3) Computational Consumption: The deployed system needs to meet the requirement of real-time processing in practice. Here, we compare our method with DLoc. Our model is approximately one-third the size of DLoc, making it more lightweight. For inference time calculation, we conduct experiments on a single NVIDIA V100 GPU. Under the same data, we evaluate 18,000 data points and take the average of the total inference time to obtain the inference time per data point. As shown in Table VIII, our method's inference time is shorter than that of DLoc.

VII. LARGE-SCALE EXPERIMENT

To further investigate the method's performance, we conduct experiments across larger scenarios. The experimental scenarios' description is shown in Table IX and the layouts are shown in Fig. 13.

We train the model in both small and large environments and then test it in both large and small environments. The results are shown in Fig. 14. As can be seen, whether transferring from the small-scale scenario to the large-scale scenario or from the large-scale scenario to the small-scale scenario, the proposed algorithm consistently outperforms the existing DLoc and SpotFi.

VIII. CONCLUSION

This paper studied the limitations of signal processing-based localization methods and the lack of cross-domain capability of existing deep learning-based approaches. By designing a deep learning-based localization approach through the perspective of signal processing, the proposed method significantly outperformed signal processing-based state-of-the-art methods. Compared to existing deep learning-based methods, we reduced the 90th percentile error of localization by up to 49%, exhibiting superior performance under cross-domain conditions. Furthermore, we released the datasets to the community for further research. Our research contributed to the advancement of WiFi-based localization and highlighted the potential of integrating signal processing with deep learning techniques for improved accuracy and generalization of indoor localization applications.

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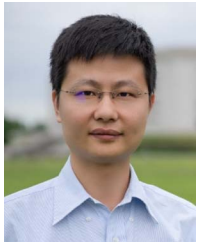

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